Assignments

REAL VARIABLES I

Fall 2026

Deadlines will appear in your Blackboard Calendar.

Weekly written homework

I will post written homework here approximately weekly. I will normally post solutions within one week of the submission deadline.

Submit through Gradescope using the links in Blackboard > Assignments > Homework.

Assignment	Due	Solutions
Homework 01	Sunday, August 30, 2026 19:00	Solutions
Homework 02	Sunday, September 6, 2026 19:00	Solutions
Homework 03	Sunday, September 13, 2026 19:00	Solutions
Homework 04	Sunday, September 20, 2026 19:00	Solutions
Homework 05	Sunday, September 27, 2026 19:00	—
Homework 06	Sunday, October 4, 2026 19:00	—
Homework 07	Sunday, October 11, 2026 19:00	—
Homework 08	Sunday, November 1, 2026 19:00	—
Homework 09	Sunday, November 8, 2026 19:00	—

- You are encouraged to work in groups of up to three.
- Make one submission per group and list every group member clearly.
- Start each exercise on a new page.
- Scan or export the complete assignment as a legible PDF.
- In Gradescope, match every exercise to the pages containing its solution.
- Write in complete sentences, state the results you use, and make the logical structure of each proof clear.

Exams

Oral exam 1

Oral exam 1 is an individual exam at the board. When you arrive, you are given a choice between two topics. You select one, prepare for up to 30 minutes without notes or other resources, and then discuss the topic at the board for about 30 minutes. The full format, grading rubric, and list of topics are on the Oral Exam 1 Information page.

The sign-up link for oral exam appointments is on Blackboard. **Please schedule your appointment so that your oral exam is done by October 27.**

Midterm 1

Midterm 1 is a timed, take-home written exam. It is available on Gradescope, accessible through Blackboard. **The deadline for completing the midterm is October 21.**

- The exam has 5 problems; choose exactly 4 and solve them. Only 4 attempted solutions will be graded, so clearly indicate which problems you chose. All problems have the same weight.
- You have 120 minutes from the moment you download the exam to solve the problems, plus a 15-minute grace period to photograph or scan your work and upload it to Gradescope. Start early enough to finish before the deadline.
- Solutions must be handwritten. The exam is closed book: no notes, books, or external help of any kind.

Online questions

Online questions usually follow each lecture and are due at the start of our next class meeting. They can be accessed through Blackboard > Assignments > Online Questions and the deadlines will appear in your Blackboard Calendar.

Online questions give you a quick chance to practice vocabulary and core skills before we build on the material. Most questions allow multiple attempts.

Academic integrity, collaboration, and artificial intelligence

You are encouraged to work with classmates, tutors, and me: talking through ideas, definitions, and problem-solving strategies is a valuable part of learning. You must understand every solution you submit and write it based on your own understanding. Please do not copy another person's solution or write-up.

You may use generative artificial intelligence as a study aid, but not to produce work that you submit for credit. For example, it is fine to ask for another explanation of the difference between pointwise and L^p convergence. It is not acceptable to ask an AI system to solve, outline, or check a specific assigned problem. **Rephrasing a generated solution in your own words is not acceptable.** If you are unsure whether a particular use is allowed, ask me before submitting.

When you submit an assignment, please disclose any collaboration or use of external tools, even when it is permitted.